In [ ]:
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EXPERIMENT 5
DATA VISUALIZATION WITH MATPLOTLIB

PIN Number:`241UDS0103`

Aim: To visualize and analyze an urban water-management dataset using
Matplotlib plots.
```

In [ ]:
```
## Step 1: Import the required libraries and dataset
```

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Read the urban water dataset
df_241UDS0103 = pd.read_csv("Experiment_5_241UDS0103_Urban_Water_Data.csv")

df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 17 entries, 0 to 16
Data columns (total 13 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Year                          17 non-null     int64
 1   Water_Production_MLD          17 non-null     int64
 2   Water_Consumption_MLD         17 non-null     int64
 3   Groundwater_Supply_MLD        17 non-null     int64
 4   Water_Tariff_INR_per_KL       17 non-null     float64
 5   Rainfall_mm                   17 non-null     int64
 6   Leakage_Rate_Percent          17 non-null     float64
 7   Treatment_Cost_Lakh           17 non-null     int64
 8   Households_Served_Thousand    17 non-null     int64
 9   Water_Demand_Growth_Percent   17 non-null     float64
 10  Production_per_Household_LPD  17 non-null     int64
 11  Tariff_to_Cost_Ratio          17 non-null     float64
 12  Cost_per_MLD                  17 non-null     float64
dtypes: float64(5), int64(8)
memory usage: 1.9 KB
```

In [ ]:
```
## Step 2: Display the dataset
```

In [2]:
```
df_241UDS0103.head()
```

Out[2]:

Columns 1–8 of 14

	Year	Water_Production_MLD	Water_Consumption_MLD	Groundwater_Supply_MLD	Water_Tariff_INR_per_KL	Rainfall_mm	Leakage_Rate_Percent
0	2010	412	365	142	4.5	912	24.5
1	2011	425	377	145	4.5	875	23.8
2	2012	438	389	148	4.8	940	23.1
3	2013	446	401	151	5.0	988	22.6
4	2014	461	416	154	5.0	901	21.9

Columns 9–14 of 14 · first column repeated

	Treatment_Cost_Lakh	Households_Served_Thousand	Water_Demand_Growth_Percent	Production_per_Household_LPD	Tariff_to_Cost_Ratio	Cost_per_MLD
0	82	185	2.8	608	0.55	0.199
1	85	192	3.1	603	0.54	0.200
2	88	199	3.2	598	0.55	0.201
3	91	207	3.4	590	0.56	0.204
4	95	215	3.7	584	0.55	0.206

In [3]:

```
df_241UDS0103
```

Out[3]: Columns 1–8 of 14

	Year	Water_Production_MLD	Water_Consumption_MLD	Groundwater_Supply_MLD	Water_Tariff_INR_per_KL	Rainfall_mm	Leakage_Rate_Percent
0	2010	412	365	142	4.5	912	24.5
1	2011	425	377	145	4.5	875	23.8
2	2012	438	389	148	4.8	940	23.1
3	2013	446	401	151	5.0	988	22.6
4	2014	461	416	154	5.0	901	21.9
5	2015	478	432	158	5.5	1035	21.2
6	2016	492	447	161	5.5	1088	20.4
7	2017	510	465	165	6.0	965	19.8
8	2018	526	481	168	6.0	1012	19.1
9	2019	541	497	172	6.5	948	18.5
10	2020	558	512	176	6.5	1104	17.9
11	2021	574	529	179	7.0	1068	17.2
12	2022	589	543	183	7.5	990	16.8
13	2023	603	559	186	7.5	1145	16.1
14	2024	621	575	190	8.0	1072	15.5
15	2025	638	592	194	8.5	1188	14.9
16	2026	655	608	197	9.0	1126	14.2

Columns 9–14 of 14 · first column repeated

	Treatment_Cost_Lakh	Households_Served_Thousand	Water_Demand_Growth_Percent	Production_per_Household_LPD	Tariff_to_Cost_Ratio	Cost_per_MLD
0	82	185	2.8	608	0.55	0.199
1	85	192	3.1	603	0.54	0.200
2	88	199	3.2	598	0.55	0.201
3	91	207	3.4	590	0.56	0.204
4	95	215	3.7	584	0.55	0.206
5	99	224	3.9	578	0.56	0.207
6	103	233	3.5	573	0.55	0.209
7	108	243	4.1	568	0.56	0.212
8	112	253	3.8	563	0.55	0.213
9	117	264	3.3	557	0.56	0.216
10	122	275	3.0	553	0.55	0.219
11	127	287	3.4	548	0.56	0.221
12	132	299	2.7	544	0.57	0.224
13	138	312	2.9	539	0.56	0.229
14	144	325	2.8	535	0.57	0.232

	Treatment_Cost_Lakh	Households_Served_Thousand	Water_Demand_Growth_Percent	Production_per_Household_LPD	Tariff_to_Cost_Ratio	Cost_per_MLD
15	150	339	3.0	530	0.58	0.235
16	157	353	2.7	525	0.59	0.240

In [4]:
```
df_241UDS0103.columns
```

Out[4]:
```
Index(['Year', 'Water_Production_MLD', 'Water_Consumption_MLD',
       'Groundwater_Supply_MLD', 'Water_Tariff_INR_per_KL', 'Rainfall_mm',
       'Leakage_Rate_Percent', 'Treatment_Cost_Lakh',
       'Households_Served_Thousand', 'Water_Demand_Growth_Percent',
       'Production_per_Household_LPD', 'Tariff_to_Cost_Ratio',
'Cost_per_MLD'],
      dtype='str')
```

In [5]:
```
# Convert Year into a numeric format
df_241UDS0103["Year"] = pd.to_numeric(df_241UDS0103["Year"])
```

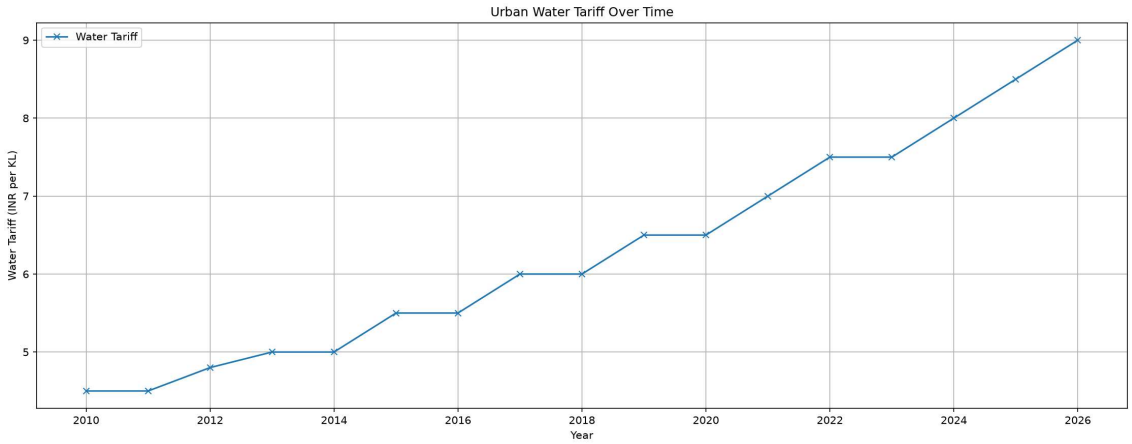
In [ ]:
```
## Step 3: Visualize Data Using Various Plots
```

In [6]:

```
plt.figure(figsize=(15, 6), dpi=150)

plt.plot(
    df_241UDS0103["Year"],
    df_241UDS0103["Water_Tariff_INR_per_KL"],
    marker="x",
    label="Water Tariff"
)

plt.title("Urban Water Tariff Over Time")
plt.xlabel("Year")
plt.ylabel("Water Tariff (INR per KL)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



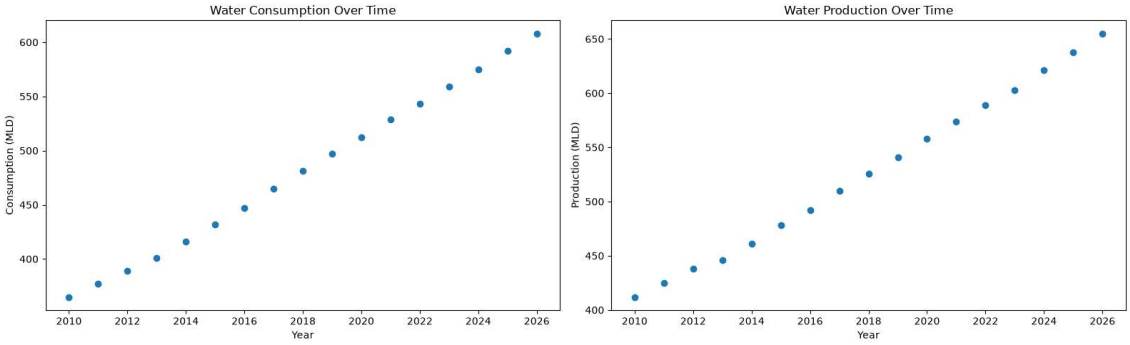
In [7]:

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fig, axes = plt.subplots(1, 2, figsize=(16, 5))

# Water consumption over time
axes[0].scatter(
    df_241UDS0103["Year"],
    df_241UDS0103["Water_Consumption_MLD"]
)
axes[0].set_title("Water Consumption Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Consumption (MLD)")

# Water production over time
axes[1].scatter(
    df_241UDS0103["Year"],
    df_241UDS0103["Water_Production_MLD"]
)
axes[1].set_title("Water Production Over Time")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Production (MLD)")

plt.tight_layout()
plt.show()
```

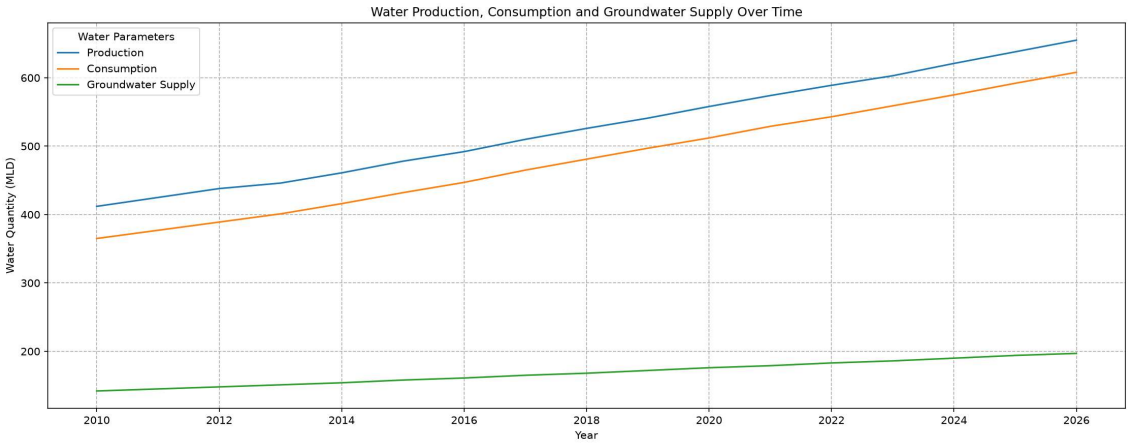


In [8]:

```
plt.figure(figsize=(15, 6), dpi=150)

plt.plot(df_241UDS0103["Year"], df_241UDS0103["Water_Production_MLD"],
label="Production")
plt.plot(df_241UDS0103["Year"], df_241UDS0103["Water_Consumption_MLD"],
label="Consumption")
plt.plot(df_241UDS0103["Year"], df_241UDS0103["Groundwater_Supply_MLD"],
label="Groundwater Supply")

plt.title("Water Production, Consumption and Groundwater Supply Over Time")
plt.xlabel("Year")
plt.ylabel("Water Quantity (MLD)")
plt.tight_layout()
plt.legend(loc="upper left", title="Water Parameters")
plt.grid(True, linestyle="--")
plt.show()
```

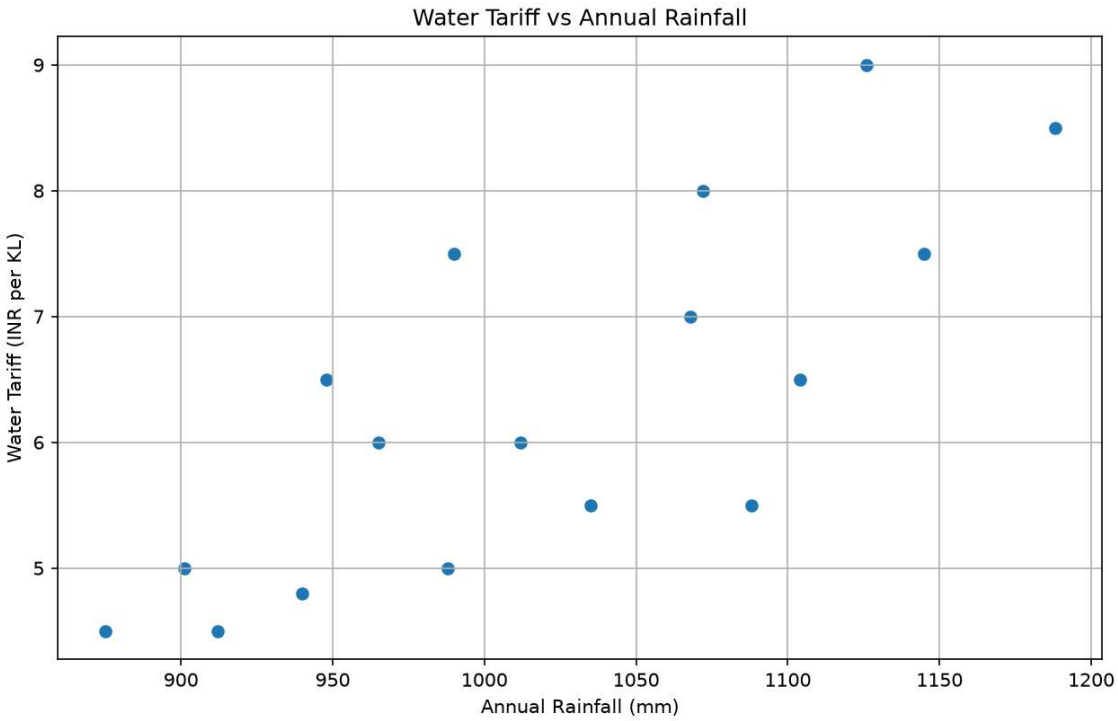


In [9]:

```
plt.figure(figsize=(10, 6), dpi=150)

plt.scatter(
    df_241UDS0103["Rainfall_mm"],
    df_241UDS0103["Water_Tariff_INR_per_KL"]
)

plt.title("Water Tariff vs Annual Rainfall")
plt.xlabel("Annual Rainfall (mm)")
plt.ylabel("Water Tariff (INR per KL)")
plt.grid(True)
plt.show()
```

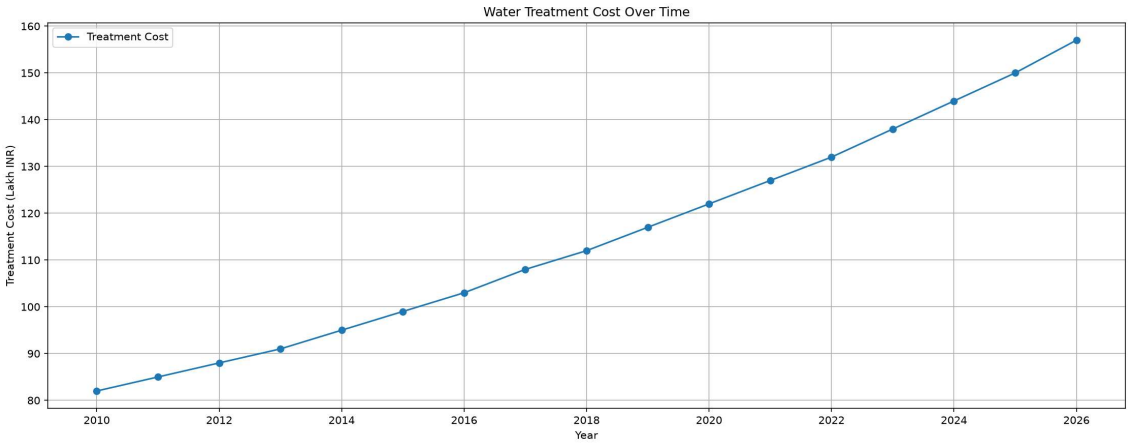


In [10]:

```
plt.figure(figsize=(15, 6), dpi=150)

plt.plot(
    df_241UDS0103["Year"],
    df_241UDS0103["Treatment_Cost_Lakh"],
    marker="o",
    label="Treatment Cost"
)

plt.title("Water Treatment Cost Over Time")
plt.xlabel("Year")
plt.ylabel("Treatment Cost (Lakh INR)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



In [ ]:

```
## RESULT
Various data-visualization techniques using Matplotlib were applied to
analyze urban water production, consumption, groundwater supply, tariffs,
rainfall, and treatment-cost trends.
```